

ZEMETCHINO, Russian Federation

WMO: 278570

Lat: **53.483N**Lon: **42.633E**Elev: **430**StdP: **14.47**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.1	-11.4	-21.7	1.7	-16.9	-15.7	2.4	-11.0	20.0	23.3	18.1	25.2	2.5	150	0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	89.9	69.7	85.9	67.4	82.6	66.2	71.8	85.7	69.6	82.3	67.7	79.1	6.1	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.9	100.6	78.6	65.0	94.1	75.7	63.4	88.8	73.6	35.8	86.2	33.8	82.3	32.3	79.1	80.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.7	14.8	12.6	-23.5	93.7	5.9	4.3	-27.7	96.8	-31.2	99.3	-34.5	101.7	-38.7	104.8		
(5)			-23.7	73.9	5.8	2.3	-27.9	75.6	-31.3	76.9	-34.5	78.2	-38.8	79.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.5	15.9	17.1	26.4	44.3	57.7	64.6	68.7	64.8	54.6	42.9	30.4	20.7
	DBStd	21.19	13.49	12.98	9.44	9.20	8.72	6.97	6.41	7.34	7.55	7.96	9.80	12.34
	HDD50	4737	1057	921	732	219	27	1	0	0	38	246	588	908
	HDD65	8576	1522	1341	1197	620	257	92	35	94	321	686	1038	1373
	CDD50	1994	0	0	0	49	267	438	580	460	175	25	0	0
	CDD65	359	0	0	0	0	32	79	150	89	9	0	0	0
	CDH74	3949	0	0	0	15	419	860	1536	992	127	0	0	0
	CDH80	1360	0	0	0	1	111	275	556	395	22	0	0	0
Wind	WSAvg	5.5	6.2	6.5	6.1	5.9	5.7	4.8	4.2	4.3	4.5	5.5	6.1	6.3
Precipitation	PrecAvg	19.5	1.3	1.1	1.1	1.1	1.5	2.1	2.9	1.9	1.8	1.7	1.7	1.5
	PrecMax	29.7	2.7	2.8	2.9	2.6	4.5	6.7	14.3	4.5	4.2	3.4	3.8	3.4
	PrecMin	11.5	0.2	0.0	0.0	0.3	0.1	0.3	0.2	0.0	0.1	0.0	0.2	0.4
	PrecStd	4.1	0.7	0.7	0.7	0.7	1.2	1.5	2.6	1.1	1.0	0.9	0.9	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.5	37.5	51.4	76.2	87.7	92.5	97.1	97.9	83.0	68.3	52.8	43.8
		MCWB	35.7	35.3	42.8	57.4	65.3	71.1	74.9	75.7	63.0	56.0	49.6	42.1
	2%	DB	35.1	35.8	45.3	70.6	83.3	86.4	89.9	89.3	77.5	62.6	48.6	39.1
		MCWB	33.8	34.3	39.3	55.5	63.5	67.5	70.6	69.8	61.5	54.0	46.7	37.5
	5%	DB	33.9	34.7	40.5	65.7	79.0	83.0	85.8	83.7	72.8	58.1	45.5	36.0
		MCWB	33.0	33.5	36.4	53.0	61.9	66.2	68.5	66.9	59.5	52.0	43.7	34.7
	10%	DB	32.7	33.5	37.5	60.7	74.4	78.9	82.3	78.8	67.9	54.6	42.2	34.1
		MCWB	31.8	32.6	34.4	50.1	59.6	64.9	67.5	64.6	57.9	50.3	40.5	33.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.4	36.0	44.2	60.1	67.7	73.5	75.0	76.1	65.7	58.8	50.5	42.3
		MCDB	36.9	36.6	51.2	70.8	81.6	88.9	93.4	97.0	76.2	64.5	52.0	43.2
	2%	WB	34.2	34.7	39.8	56.5	65.1	69.6	72.3	70.8	63.4	55.4	47.0	37.7
		MCDB	34.7	35.3	44.2	68.0	79.6	82.3	86.6	85.2	73.2	59.7	48.4	38.9
	5%	WB	33.1	33.7	36.8	53.8	63.0	67.6	70.3	68.2	61.2	52.8	43.8	35.0
		MCDB	33.8	34.4	39.7	63.7	76.0	79.3	82.7	80.6	70.2	57.3	45.3	35.8
	10%	WB	31.9	32.6	35.0	51.1	60.8	65.9	68.4	66.0	58.7	50.4	40.7	33.5
		MCDB	32.9	33.5	37.0	59.8	72.9	76.7	79.6	76.2	66.7	54.1	42.1	34.0
Mean Daily Temperature Range	5% DB	MDBR	11.3	12.9	14.1	18.5	21.0	19.2	19.9	20.4	17.9	13.2	9.0	9.5
		MCDBR	7.1	8.8	15.8	26.3	27.1	26.4	26.2	28.6	27.0	20.4	11.9	7.4
		MCWBR	7.1	8.3	10.8	14.0	12.3	11.3	11.0	12.0	12.9	12.7	10.1	6.9
	5% WB	MCDBR	7.1	8.5	13.0	24.5	24.5	22.8	22.9	25.3	22.2	18.2	10.5	6.6
		MCWBR	7.2	8.3	9.6	14.0	12.3	10.8	10.4	11.7	11.8	12.6	9.7	6.7
Clear-Sky Solar Irradiance	taub	0.295	0.331	0.358	0.426	0.392	0.387	0.412	0.432	0.379	0.358	0.329	0.298	
	taud	2.097	2.027	2.043	2.091	2.313	2.353	2.295	2.224	2.352	2.327	2.228	2.105	
	Ebn at Noon	203	232	255	253	272	274	264	249	250	226	191	175	
	Edh at Noon	23	34	41	45	38	37	38	39	30	25	20	19	
All-Sky Solar Radiation	RadAvg	265	556	1057	1344	1778	1833	1850	1519	972	480	200	147	
	RadStd	52	96	140	134	155	176	169	137	128	69	47	45	